

Milestone 1 Evaluation Document

Project: Home Network Solutions

Team Members:

- Bryan Bermudez (bbermudez2023@my.fit.edu)
- Joshua Revels (jrevels2023@my.fit.edu)
- Brennan Ramsey (bramsey2023@my.fit.edu)
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Faculty Advisor & Client: Dr. Mohan, FIT Department of EE and CS

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Milestone 1 Progress Matrix:

Task	Completion %	Joshua	Brennan	Bryan	Nick	To-do
Requirement Document	100%	20%	20%	40%	20%	None
Design Document	100%	25%	25%	25%	25%	None
Test Plan	100%	15%	15%	10%	60%	None
Investigate tools	100%	25%	25%	25%	25%	None
Set up infrastructure	65%	50%	15%	0%	0%	Jira

Task Discussion:

1. Requirement Document:
 - We defined what the project is, what it does, and what needs to be done to start building it. From this, we discovered the limitations of using third-party software and how to plan around these constraints.
2. Design Document:
 - We defined the project's backend, providing details like how features will be implemented and which tools or software will power them. From this, our team was able

to build a solid understanding of the communication flow between services within each device.

3. Test Plan:

- We determined that unit, incremental, system, and usability testing will be vital to successfully accomplishing the project's goals on schedule. Unit testing will allow us to test important features before the main merge, incremental testing will keep the project on time by adding completed features piece by piece, system testing will assure that the database and the file system are in sync, and usability testing will assure that our project is usable by non-technical users.

4. Investigate tools:

- The two extra tools that we decided on using are SQLite and Caddy. SQLite was chosen due to the limited data needs of this product, so a simple, lightweight version of SQL is optimal. Caddy was chosen to handle the webserver over Apache, as multiple team members have experience with it.

5. Set up infrastructure:

- The test server was successfully set up, and team members can connect to the server from anywhere. The biggest obstacle for this was allowing SSH connections with devices outside the local network, but this was fixed by preemptively installing Tailscale onto the server to allow team members to tunnel into the server.
- The Jira project board was successfully set up for a Kanban structure, as these task statuses are similar in flow to the milestone task matrices. However, work still needs to be done to import tasks and stories in order for Jira to become a part of the team's workflow.

Team Member Discussion:

Joshua Revels:

- Worked through various parts of the requirement and design documents. Reviewed test and design documents. Set up the test server that will be used for the remainder of the project. Did research on which webserver handler to use.

Brennan Ramsey:

- Created consistent formatting and layout in requirement, design, and test documents. Researched tools and the system design of some features; primarily file sync, VPN, and the helper program. Worked on various sections within each document. Set up a Jira board with Kanban structure.

Bryan Bermudez:

- Focused primarily on requirement and design documents. Created the base template for all documents and initial planning for test, design, and requirements documentation. Researched into tools, specifically Home Assistant and how it would be utilized in the project.

Nick Bleeker:

- Focused on the requirements and test documents, specifically the overall test strategy. Also researched which tools to implement and how each one fits into the project. Additionally, researched existing home lab setups to find features that are commonly used and which ones tend to cause issues. This research helped find which features are worth prioritizing in the project's design.

Milestone 1 Client/Faculty Advisor Meetings:

1. 8/26/26 - 12:20 PM
2. 9/23/26 - 12 PM

Client/Faculty Advisor Feedback:

- See Faculty Advisor Feedback below.

Milestone 1 Faculty Advisor Feedback

1. Requirements Document

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2. Design Document

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3. Test Plan

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4. Investigate tools

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5. Set up infrastructure

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Milestone 2 Task Matrix:

Task	Joshua	Brennan	Bryan	Nick
Finish setting up infrastructure	10%	55%	25 %	10%
Install each tool with tests	30%	25%	25 %	20%
Create an install script	25%	25%	25 %	25%
Create login page	15%	20%	15 %	50%

Milestone 2 Task Discussion:

1.
 - We will finish importing tasks and requirements to the Jira board. For development purposes, we will ensure the team's SSH connections are run over Tailscale for security and ease of use when remote. We will finish user management within Ubuntu, so each team member and service has its own user with specific permissions, enhancing security.
2.
 - Pi-hole, Jellyfin, Home Assistant, Caddy, Samba, and Git will be installed and tested individually on the server. Testing each one on its own ensures that each application works as expected and is configured correctly.
3.
 - Once each application is installed and tested, the processes will be turned into a single script that is used to set up the server from scratch. This step is essential for making a fresh install easy for non-technical users.
4.
 - Start working on the web app with the goal of creating a log in page that loads into a temporary dashboard. This gives us something to build off of for Milestone 3.

Faculty Advisor Signature: _____

Date: _____